PROJECT MANAGEMENT DOCS

PROJECT BRIEF

LITPATH AI: SMART PATHFINDER FOR THESES AND DISSERTATIONS

ASIA PACIFIC COLLEGE
3 HUMABON, MAGALLANES
MAKATI CITY, METRO MANILA 1232

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PROJECT NAME	LitPath AI: Smart Pathfinder for Theses and Dissertations		
PROJECT MANAGER	Tracie Tomon	EMAIL	tctomon@student.apc.edu.ph
START DATE END DATE	March 2025 June-July 2026	BUDGET	
PROJECT OVERVIEW	The DOST-STII Library houses a valuable and growing collection of graduate theses and dissertations from DOST Scholars, cataloged through its Online Public Access Catalog (OPAC). Despite the value of this collection, it remains underutilized due to critical limitations in the OPAC's search functionality, including exact keyword matching, slow processing times, and a lack of Artificial Intelligence. These limitations create a significant barrier to knowledge discovery for students, academic researchers, and DOST employees. Users who are exploring new research areas or are unsure of precise terminology find it difficult to discover relevant materials, leading to research inefficiencies and preventing the library from maximizing the impact of its key academic resources. Specifically, the key challenges with the current system are: 1. Limited Search Mechanism: The current system relies heavily on exact keyword and title-based searches, lacking AI-powered search capabilities to interpret a user's research interests or intent. This makes the system ineffective for users exploring broader topics or those unfamiliar with precise terminology, often yielding results that are irrelevant or incomplete. 2. Poor Relevance and Efficiency: The OPAC struggles to consistently generate accurate results due to the absence of advanced, intent-driven search algorithms. Internal assessments reveal that fewer than 7 out of 10 results match a user's actual research interests. Consequently, users are forced to spend approximately 15 minutes per session manually reviewing and sifting through irrelevant results to find useful materials. 3. Lengthy Searching Performance: The search function exhibits severe performance issues, with search queries taking over a minute to return results. This processing time is significantly slower than the acceptable industry standard of 30 seconds, leading to poor user experience and discouraging further exploration of the platform. 4. Lack of Essential Research Support Features: The current system lacks key built-in tools essential for modern academic research, specifically		

	automated citation generation and bookmarking capabilities. As a result, users must manually compile citations and manage their research materials independently, which increases the time spent on administrative tasks and raises the risk of formatting errors or disorganized references.
OBJECTIVES	The primary goal of this project is to develop and implement an AI-powered search application to increase the discovery and use of the DOST-STII Library's collection of theses and dissertations. The project will achieve this through the following specific objectives: 1. To improve search efficiency and precision, the system will directly solve the limitations of the current exact-match search mechanism. Success will be measured by achieving a user satisfaction score of at least 3.5 out of 5 from testers on the relevance and ease of use of the new search features. 2. To help users discover more relevant studies, the system will leverage AI-powered algorithms to provide semantically relevant search results. These algorithms will analyze user queries and content to ensure that the top 10 search suggestions have a minimum relevance accuracy of 70%, based on user feedback and expert validation. 3. To create a better and faster user experience, the interface will be designed to be intuitive and responsive, with the goal of returning results significantly faster compared to the existing system, resulting in a more efficient user experience. 4. To make citing sources easier for researchers, the system will feature a built-in automated citation generator that supports multiple citation styles, including APA 7th Edition, MLA 9th Edition, Chicago Manual of Style, and IEEE. Accuracy will be verified through expert review and user testing to ensure correct citation formatting.
SCOPE	The scope of the LitPath AI project includes the delivery of the following tangible outputs: 1. Standalone Web Application: A fully functional, independent user interface that operates separately from the existing DOST-STII OPAC. 2. AI-Powered Semantic Search Engine: A backend search algorithm designed to interpret the meaning, context, and research intent behind user queries, moving beyond traditional exact-match keyword searching.

	3. Client-Provided Database: An independent backend database populated exclusively with the sample thesis and dissertation papers provided directly by DOST-STII for the system's search repository. 4. Automated Citation Generator: A built-in tool that automatically formats citations for selected research materials in APA, MLA, Chicago Manual of Style, and IEEE formats. 5. Research Management & Feedback Tools: A bookmarking feature allowing users to save and organize their discovered research materials, alongside an integrated feedback tool to capture user ratings and comments. 6. Admin Analytics Dashboard: A dedicated dashboard providing DOST-STII library administrators with visibility into search trends, usage patterns, and user interactions.
DELIVERABLES	1. Working Prototype Ready for Deployment: A fully functional, standalone web application featuring an AI-powered semantic search engine capable of analyzing user research intent. This complete system includes a responsive user interface, a built-in automated citation generator (supporting APA 7th, MLA 9th, Chicago, and IEEE styles), a bookmarking feature for saving and organizing research materials, an admin dashboard for tracking search trends and usage patterns, and an integrated user feedback tool to measure satisfaction and validate search relevance. 2. Comprehensive Project Documentation: A complete package containing all project documents required for formal handover, future use, and maintenance. This includes the technical manual, a user guide for researchers and library staff, and a final report verifying that all project objectives have been successfully met.
SUCCESS CRITERIA	1. The project delivers a 100% working application with all defined features fully complete and operational. 2. System achieves an average user satisfaction score of at least 3.5 out of 5 from the test group of students and DOST employees. Feedback must also confirm that the system makes their research process feel faster and easier. 3. The new search system is successfully deployed as a standalone

	web application via Proxmox and consistently processes queries and displays results faster than the traditional library OPAC system. 4. All key deliverables - the AI engine, the user interface including the citation tool, user feedback system, and all documentation, have been formally signed off on and accepted by the project stakeholders.
TARGET AUDIENCE	The project is intended for multiple audiences: 1. Students / Researchers – Individuals conducting research for academic papers or projects. 2. System Admin – DOST-STII administrators and facilitators who use the system to assist library users during reference interviews and understand usage analytics.

SPONSOR ACCEPTANCE

Approved by the Project Sponsor:

_____ Date: April 1, 2026

Khasian Eunice Romulo
Science Research Specialist II / Unit Head
(Digital Information and Processing Delivery Unit)